

Website LinkedIn

CORE COMPETENCIES

Strategic research leadership
Malaria vaccine research and biomedical AI delivery.

Academic and industry partnerships
Mabtech, Muhimbili University and CIDEIM collaborations.

Research grant acquisition
Competitive foundation awards and public funding contributions.

Team leadership and capability development
MSc and PhD supervision; expert onboarding and calibration.

Operational execution and governance
Milestones, workload allocation, acceptance criteria and SOPs.

CITIZENSHIP AND LANGUAGES

Swedish and Colombian citizen

- Spanish - native speaker
- English - bilingual proficiency
- Swedish - working proficiency
- German - elementary proficiency

David F. Plaza, PhD

Sickla Strand 71, 13134 Nacka, Sweden
+46 73 573 92 83 | dfplazag@gmail.com
dfplazag.github.io/david-plaza-site/
linkedin.com/in/david-f-plaza-phd-43a06277

SUMMARY

Biomedical research leader and ETH Zurich PhD combining academic partnerships, team development and operational execution with biomedical AI delivery. Secured SEK 700,000 from private foundations as main applicant; major contributor to SEK 9,299,000 in public research funding in 2021. Brings international collaboration, translational science, grant acquisition and multidisciplinary project leadership.

PROFESSIONAL EXPERIENCE

Strategic Project Lead

Lumos AI | Genomics | Paris, France

Sep 2026 - Present

- Lead the construction of reinforcement learning environments for biomedical AI with Research and Engineering, specifying scientific tasks, tools, verifiers and acceptance criteria.
- Own expert-data delivery from scoping and production planning to contributor onboarding, workload allocation, milestone tracking and final acceptance.
- Develop reference solutions, scoring rubrics and SOPs; calibrate contributors, adjudicate expert submissions and communicate scientific quality decisions.

Assistant Professor

Karolinska Institutet | Stockholm, Sweden

Sep 2021 - Aug 2025

- Led translational malaria vaccine research connecting genomic surveillance, AI-assisted structural modelling and antibody profiling.
- Led an academic-industry partnership with Mabtech on recombinant antigens and antibody responses, resulting in a joint *Frontiers in Immunology* publication.
- Secured SEK 700,000 from Clas Groschinskys Minnesfond and Tore Nilsons Stiftelse as main applicant; contributed substantially to two Swedish Research Council project grants totalling SEK 9,299,000 for the Division of Infectious Diseases in 2021.
- Supervised MSc projects, co-supervised doctoral research and mentored visiting trainees; defined objectives, reviewed progress and supported scientific writing.
- Supported research governance through scientific integrity guidance and a divisional lecture on authorship and responsible data management.

EDUCATION

PhD Microbiology and Immunology

ETH Zurich, Switzerland
2014

MSc Biology

Immunology and Infection Biology
Uppsala University, Sweden
2010

BSc Biology

Universidad Nacional de Colombia
2006

TRAINING AND CERTIFICATES

Google Project Management Professional Certificate

Google via Coursera | Completed
Project planning, risk and quality management, stakeholder communication, Agile and Scrum.

Life Science Entrepreneurship and Innovation

SmiLe Venture Hub | 2022

International Vaccinology Course

International Vaccine Institute | 2022

Doctoral Supervision

Karolinska Institutet | 2023

CAREER BREAK

Parental leave
Sep 2025 - Aug 2026
Stockholm, Sweden

PROFESSIONAL EXPERIENCE

Postdoctoral Scientist

Karolinska Institutet | Stockholm, Sweden

Sep 2019 - Aug 2021

- Developed a PacBio long-read platform for malaria antigen surveillance and discovery; validation included 282 *Plasmodium falciparum* samples from Tanzania.
- Connected molecular epidemiology with memory B-cell and antibody research through multidisciplinary collaboration.

Postdoctoral Fellow

FIDIC | Bogotá, Colombia

Apr 2018 - Aug 2019

- Led a small research team investigating host receptors for malaria invasion through peptide-mediated affinity capture and functional characterization; produced a joint *Journal of Proteomics* publication with Temple University collaborators.
- Developed NHP-Immunome using RNA-seq datasets from four non-human primate species to inform translational model selection.

Postdoctoral Fellow

Harvard University | Cambridge, USA

Nov 2014 - Nov 2017

- Investigated chromatin regulation and embryonic development in *Caenorhabditis elegans*, combining biochemical experiments, mass spectrometry and computational analysis.

Research and Teaching Assistant

ETH Zurich | Zurich, Switzerland

Jun 2010 - Jun 2014

- Combined comparative transcriptomics and protein characterization to study fungal defence mechanisms; supervised two MSc projects.

Research Assistant

FIDIC | Bogotá, Colombia

Jul 2006 - Jun 2008

- Characterized peptide-based tuberculosis vaccine candidates and developed a flow cytometry method to quantify host-cell invasion.

SELECTED PUBLICATIONS

2025 | *STAR Protocols* 6(4):104093.
Protocol for genomic surveillance of *Plasmodium falciparum* antigens using amplicon-based PacBio long-read sequencing. Sole author.

2023 | *Cell Reports Methods* 3(9):100574.
A genomic platform for surveillance and antigen discovery in *Plasmodium* spp. using long-read amplicon sequencing. First and corresponding author.

2024 | *Frontiers in Immunology* 15:1501700.
Naturally acquired IgG responses to *Plasmodium falciparum* do not target the conserved termini of the malaria vaccine candidate Merozoite Surface Protein 2. Last and corresponding author.